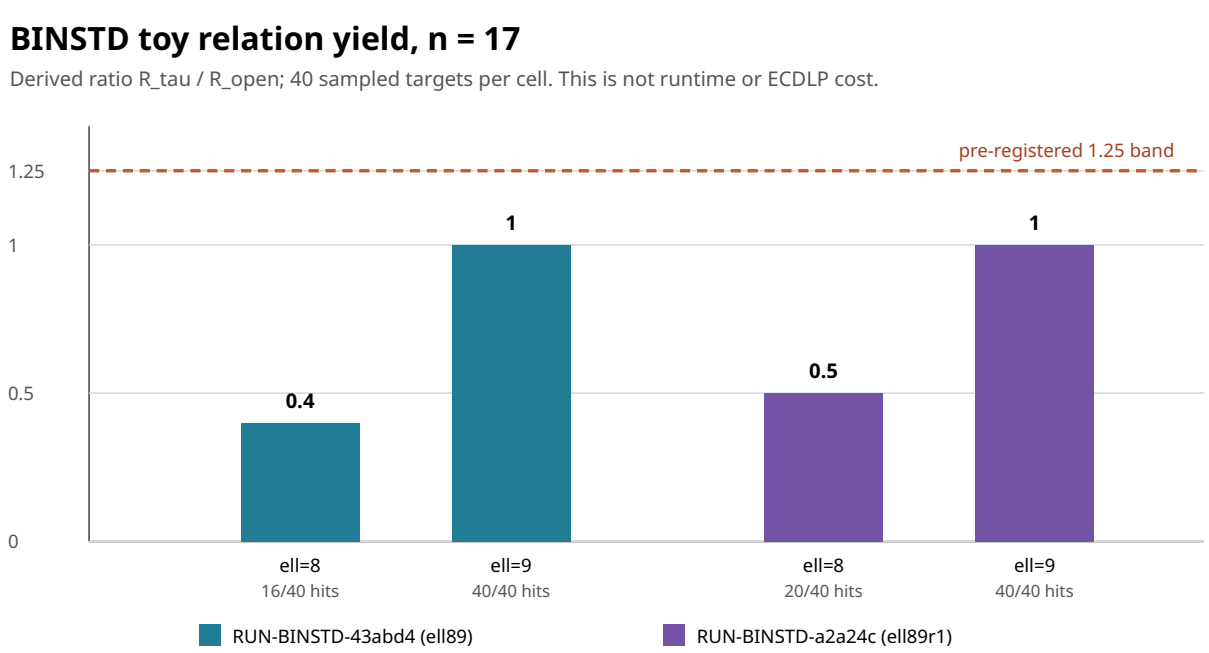
Source-pinned replay of BINSTD ell89 toy panels

2026-10-22 EXP-BINSTD-8bdf86 H-BINSTD-5a212b

This validation replay used the historical code commits recorded in the two run environments and copied their hash-matched Stage-0 freeze files into isolated temporary directories. It did not write to any frozen run record. This is not a new experiment or evidence decision.

Panel / Stage-1 run	ℓ	Tau hits	Open hits	R_τ/R_{open}	Band
ell89 / RUN-BINSTD-43abd4	8	16/40	40/40	0.4	1.25
ell89 / RUN-BINSTD-43abd4	9	40/40	40/40	1.0	1.25
ell89r1 / RUN-BINSTD-a2a24c	8	20/40	40/40	0.5	1.25
ell89r1 / RUN-BINSTD-a2a24c	9	40/40	40/40	1.0	1.25



Both freshly generated yield matrices, admissible-dimension files, and results notes match the archived bytes exactly. Raw-result payloads match after excluding the new timestamps. The source files, dated specifications, and Stage-0 freeze artifacts match their recorded SHA-256 pins. The committed `receipt.json` lists all hashes; a second end-to-end invocation of the committed replay command reproduced it byte for byte. The first local replays took 355.8 and 315.6 seconds on Python 3.9.25; these are not benchmark timings. A one-count mutation in a copied matrix caused the verifier to fail before writing a receipt, including under `python3 -O`.

Scope. The four recorded toy ratios lie below the pre-registered 1.25 band at $n = 17$, $\ell \in \{8, 9\}$, under the exhaustive $m = 3$ pin. The open arm hit all 40 targets in each cell, so its observed rate was 1; the tau rate cannot exceed 1, making the 1.25 ratio unattainable on those measured cells. This is a ceiling limit on the tested protocol. An independent polynomial irreducibility calculation passed, and the degree-1 and two degree-8 factors of $x^{17} - 1$ multiplied out, giving subset sums $\{0, 1, 8, 9, 16, 17\}$. The normal-basis link to invariant subspaces has no Lean proof here. A separate binary-field implementation checked all 8^3 ordered triple sums and the pair-membership shortcut on 12 constructed hits and 12 sampled misses per pinned runner. This bounded control does not independently recompute the full panel or prove the runner in Lean. No ECDLP break, asymptotic transfer, runtime advantage, or rho comparison follows. No uncertainty interval was recorded for the 40-target hit rates.

Reproduce. Run `python3 docs/reports/2026-10-22-binstd-ell89-replay/replay.py -run` from the repository root. The detailed report and graph source are beside this PDF. The canonical performance graphs were checked; the replay creates no new speedup or attack-cost row.